

CoFiTok: COARSE-TO-FINE DENOISING TOKENS FOR PIXEL-SPACE DIFFUSION

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ABSTRACT

Pixel-space diffusion models usually predict a full dense noise field with a single monolithic network output. We introduce CoFiTok, a denoising-token factorization that represents the dense noise prediction as an ordered sum of negative-noise components. Each token is expanded by a restricted, condition-free synthesis operator that only sees the current token, so prefix sums form valid partial noise predictions and produce controllable partial denoising results in pixel space. Across eight current datasets spanning CIFAR-10, ImageNet-family 64x64 and 256x256 settings, FFHQ-64, and AFHQv2-64, CoFiTok improves cross-batch prefix path AUC over an endpoint-only factorized control on all datasets and avoids zero-token leakage that appears when the synthesis operator is made deeper. In a two-seed ImageNet-256 confirmatory study, the learned order ranks first among all 24 K4 permutations for both seeds while preserving the endpoint sum. Frechet-style generation metrics favor EDM or direct dense epsilon on most rows, so the supported claim is not unconditional generation superiority; it is that ordered restricted dense-noise factorization yields prefix-controllable denoising without relying on a VAE-style decoder.

1 INTRODUCTION

Diffusion models reverse a noising process by repeatedly predicting a dense noise field. This dense prediction is effective, but it is also monolithic: there is no direct way to ask how much of the denoising update has been explained, whether the update has an interpretable coarse-to-fine order, or how to stop after a valid partial prediction. Ordered visual tokenizers and diffusion-autoregressive methods suggest that generation can benefit from structured intermediate representations. The question in this paper is narrower: can the dense pixel-space noise prediction itself be factorized into ordered denoising tokens?

CoFiTok rewrites the predicted noise field as

$$\hat{\epsilon}^{(K)} = \sum_{k=1}^K S_k \left(z^{(k)} \right), \quad (1)$$

where each $z^{(k)}$ is a denoising token and S_k expands only that token into one dense component. Prefixes are meaningful by construction:

$$\hat{\epsilon}^{(1:m)} = \sum_{k=1}^m S_k \left(z^{(k)} \right), \quad (2)$$

$$\hat{x}_0^{(m)} = \frac{x_t - \sigma_t \hat{\epsilon}^{(1:m)}}{\alpha_t}. \quad (3)$$

The central design constraint is that the synthesis operator must not become a decoder. In the default CoFiTok model, S_k is token-only, bias-free, shallow, local, and condition-free. It does not receive the noised image, timestep, class label, text prompt, previous tokens, skip features, or a learned constant. This restriction forces the token itself to carry the sample-specific negative-noise information and gives the zero-token test a concrete meaning: $S_k(0) = 0$ by construction.

The current evidence supports a scoped method claim. CoFiTok learns ordered denoising components across eight datasets, including ImageNet-family 64x64 and 256x256 settings and non-classification FFHQ/AFHQv2 rows. Compared with an endpoint-only factorized predictor, CoFiTok consistently improves prefix/path alignment, while matched objective, prediction, and synthesis ablations expose why ordering and restricted synthesis matter. Generated-sample distribution metrics do not show a stable unconditional generation-quality win, which clarifies the present contribution: CoFiTok makes dense noise prediction prefix-controllable without relying on a final VAE-style decoder.

Contributions.

1. We formulate dense pixel-space diffusion noise prediction as an ordered sum of denoising-token components.
2. We introduce a restricted, condition-free synthesis operator whose token-only interface and bias-free construction prevent decoder-style shortcuts and make $S_k(0) = 0$ testable.
3. We train CoFiTok with a denoise-path objective that makes token prefixes approximate a partial denoising trajectory rather than only the final clean image.
4. We evaluate prefix controllability, component ordering, non-degeneration, endpoint quality, matched short-run generation metrics, and tokenizer-related reconstruction baselines across the current eight-dataset evidence matrix.

2 RELATED WORK

Diffusion models and latent diffusion. Denoising diffusion probabilistic models learn to reverse a Gaussian noising process by predicting noise targets (Ho et al., 2020). Latent diffusion models move this process into a learned latent space and decode the final latent through a VAE-style decoder (Rombach et al., 2022). CoFiTok instead stays in the pixel-space denoising algebra and factorizes the dense noise prediction itself. The CoFiTok token is not a final image latent; it is a negative-noise component that directly contributes to the pixel-space denoising update.

Trajectory ordering and latent-pixel interaction. Latent Forcing studies how changing the schedule between latent and pixel variables changes the information revealed by the diffusion trajectory (Baade et al., 2026). CoFiTok is inspired by this view of trajectory ordering, but tokenizes the noise prediction inside the reverse trajectory rather than introducing a separate latent-pixel trajectory.

Ordered visual tokenizers. Spectral Image Tokenizer, FlexTok, Matryoshka Representation Learning, M3, and related nested-token approaches organize image or multimodal representations so that prefixes or subsets can be useful (Esteves et al., 2025; Bachmann et al., 2025; Kusupati et al., 2024; Cai et al., 2024). These methods operate on image representations or visual-token budgets. CoFiTok operates on dense diffusion noise prediction. Its prefixes are valid pixel-space denoising updates, not partial decodes from a general image tokenizer.

Diffusion-autoregressive visual tokens. Selftok and D-AR are the closest conceptual neighbors because they connect diffusion processes and autoregressive tokens (Wang et al., 2025; Gao & Shou, 2025). CoFiTok differs in the represented object and in the anti-degeneration mechanism. CoFiTok tokens are continuous negative-noise components, and the synthesis operator is deliberately weak and condition-free. The main claim is not that diffusion steps are visual tokens, but that dense noise prediction can be decomposed into ordered components whose prefixes are directly usable in the diffusion denoising formula.

3 METHOD

3.1 FORWARD NOISING AND DENSE PREDICTION

For a clean image x_0 , timestep t , and Gaussian noise ϵ , the forward process is

$$x_t = \alpha_t x_0 + \sigma_t \epsilon. \quad (4)$$

CoFiTok method overview

Dense pixel-space noise prediction is factorized into ordered restricted denoising components.

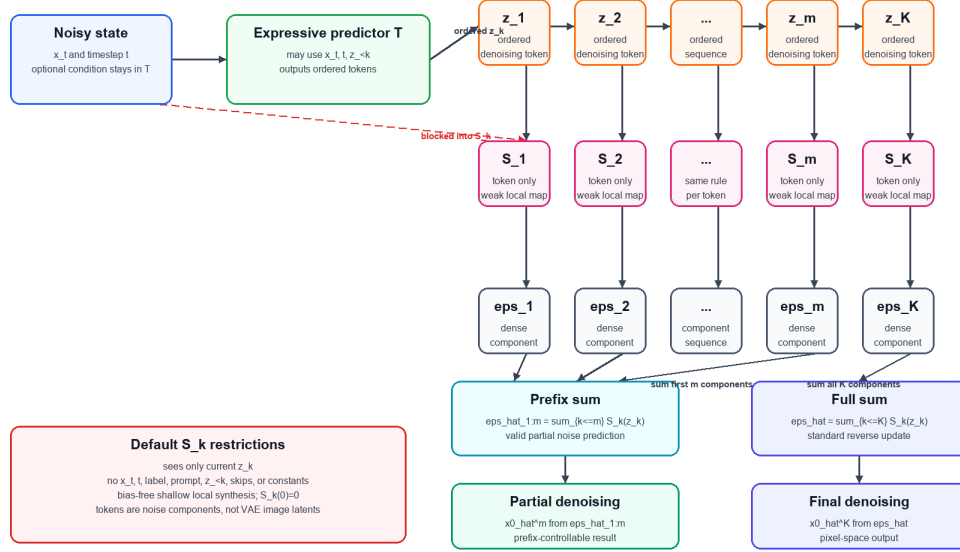


Figure 1: CoFiTok factorizes dense pixel-space diffusion noise prediction into ordered denoising tokens. The predictor may be expressive, but each synthesis operator S_k receives only the current token and is restricted to shallow, condition-free, bias-free local synthesis. Any prefix of components forms a valid partial noise prediction and therefore a valid pixel-space denoising result. Dashed blocked paths mark information that must not enter S_k , including x_t , timesteps, labels, prompts, previous tokens, skip features, and learned constants.

A conventional diffusion model predicts $\epsilon_\theta(x_t, t, c)$. CoFiTok instead predicts the ordered component sum in Eq. 1. The current implementation uses unconditional models for the main MVP evidence. Class labels may exist in datasets, but the restricted synthesis operator never receives labels, prompts, timesteps, previous tokens, or image features. Figure 1 shows the dataflow and marks the blocked paths into S_k that define the default method boundary.

3.2 NEXT DENOISING TOKENS

The token predictor T_k may be expressive. Our default implementation sees the noised image, timestep, and previous tokens:

$$z^{(1)} = T_1(x_t, t), \quad (5)$$

$$z^{(k)} = T_k(x_t, z^{(<k)}, t), \quad k > 1. \quad (6)$$

In the tiny MVP model, T_k is a small U-Net-like predictor with token heads. Autoregressive feedback is not part of the synthesis restriction or the core factorization claim: a simultaneous variant may instead use $z^{(k)} = T_k(x_t, t)$. We evaluate this choice directly.

3.3 RESTRICTED SYNTHESIS OPERATOR

Each token is expanded independently:

$$\epsilon^{(k)} = S_k(z^{(k)}). \quad (7)$$

The default S_k is a bias-free projection followed by a bias-free local convolution. It is token-only and condition-free. Because no layer has bias and there is no learned constant input, zero tokens map to zero components:

$$S_k(0) = 0. \quad (8)$$

This restriction separates CoFiTok from a learned decoder. A deep- S_k ablation is included only to test degeneration risk: it uses biased nonlinear convolutions while preserving the token-only interface. This ablation can improve apparent endpoint metrics but fails the zero-token diagnostic, so it is not the default method.

3.4 PREFIX DENOISING

For any prefix length m , CoFiTok forms the prefix prediction and pixel-space denoising result in Eq. 3. Thus every prefix is a valid partial noise prediction and every prefix produces a pixel-space denoising result. This is the primary interface exposed by the method.

3.5 DENOISE-PATH OBJECTIVE

The current MVP objective is

$$\mathcal{L} = \mathcal{L}_\epsilon + 0.15 \mathcal{L}_{\text{path prefix}} + 0.30 \mathcal{L}_{\text{path component}}. \quad (9)$$

The denoise path interpolates from the zero-update x_t reconstruction toward coarse-to-fine spatial targets. The progress schedule uses power 1.5 in the current best MVP configuration. This objective deliberately avoids forcing early prefixes to be immediately clean; instead, prefixes are trained to follow a partial denoising path.

4 EXPERIMENTS

4.1 SETUP

Datasets and models. We evaluate on CIFAR-10, Tiny ImageNet-200, ImageNet-1K 64x64 HF, strict-source downsampled ImageNet-64, FFHQ-64, AFHQv2-64, ImageNet-256 10%, and full ImageNet-256. The HF ImageNet-64 row and the strict-source downsampled ImageNet-64 row are reported separately; older HF-derived results are not silently renamed as strict-source results. All matched experiments use small unconditional models and short training budgets. The primary model is K8 light CoFiTok at 64px and K4 at 256px. We compare it with a parameter-matched direct dense predictor, an endpoint-only factorized control, explicit objective/prediction/synthesis ablations, Improved DDPM, and EDM. FlexTok/TiTok reconstruction and official D-AR/MAR/ReTok 50K results are protocol-separated secondary context.

Protocols. The P0 rows match dataset, resolution, split, and nominal optimizer steps; batch size, nominal images seen, parameter count, and NFE are reported because the external methods are not compute matched. The direct dense control is the strictest factorization control: it is parameter matched and uses the same training budget, but is not claimed to be FLOP or latency matched. The broad matrix uses locked per-dataset evaluation sizes. Repeated and ablation claims use seeds 103/139 and 1,024 validation images at fixed $t = 500$. All power-sweep models are evaluated against the same progress-1.5 path.

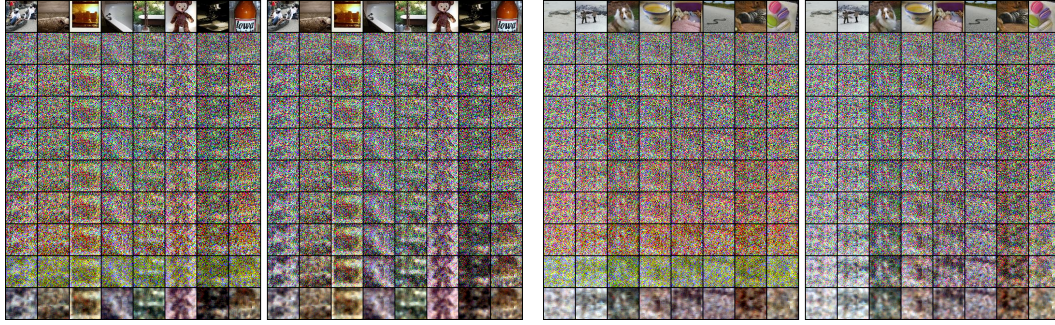
Metrics and evidence tiers. We report endpoint PSNR and MSE, path AUC, effective token count, and zero-input energy. Lower path AUC means better alignment of all prefixes to the prescribed partial-denoising path; endpoint metrics alone do not measure that interface. Frechet-style values are internal comparison metrics rather than SOTA FID claims. The locked evidence matrix contains 80 matched P0 cells, 16 tokenizer eval-only cells, and 24 explicitly protocol-blocked cross-task cells; full settings and provenance are in the appendix.

4.2 RESULTS

All-dataset evidence. Table 1 isolates the supported claim. C is CoFiTok, D is the parameter-matched direct dense predictor, and E is the endpoint-only factorized control. CoFiTok has lower path AUC than E on all eight datasets and exactly preserves the restricted zero-token contract. Its endpoint PSNR is close to, but not uniformly better than, direct dense.

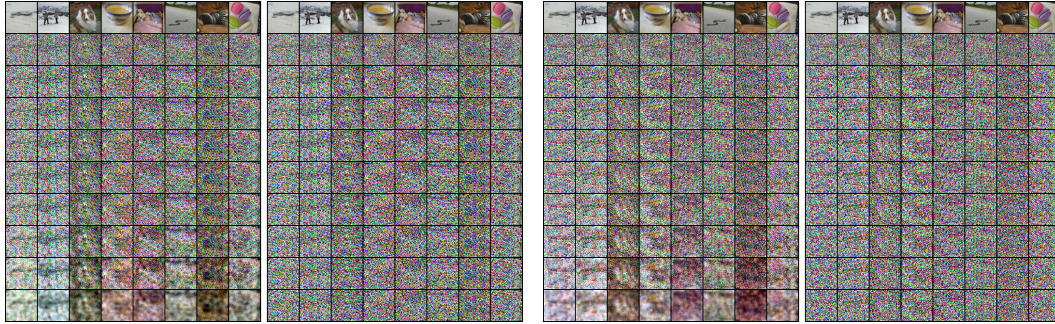
Table 1: Diagnostics for the supported CoFiTok claim. Higher PSNR is better; lower path AUC and zero ratio are better.

Dataset	C PSNR	D PSNR	C path	E path	C effK	E effK	C zero	deep- S_k zero
CIFAR-10	14.245	14.407	0.0967	0.7906	6.84	5.24	0.0000	0.0619
Tiny-IN	14.348	14.453	0.0731	1.1177	6.80	5.57	0.0000	0.0505
ImageNet-64 HF	14.897	15.004	0.0672	1.1120	6.79	5.33	0.0000	0.0522
ImageNet-64 strict	14.521	14.656	0.0705	1.0502	6.82	5.36	0.0000	0.0535
FFHQ-64	15.024	15.126	0.0676	1.0830	6.82	5.41	0.0000	0.0518
AFHQv2-64	15.167	15.154	0.0653	1.0185	6.82	5.45	0.0000	0.0561
ImageNet-256 10%	13.627	13.490	0.1416	1.5686	3.36	3.67	0.0000	0.0179
ImageNet-256	13.809	13.770	0.1325	1.5003	3.37	3.67	0.0000	0.0185



(a) Tiny-IN: endpoint-only (left), CoFiTok (right).

(b) ImageNet-64: endpoint-only (left), CoFiTok (right).



(c) Restricted S_k : ordered (left), shuffled (right).

(d) Deep S_k : ordered (left), shuffled (right).

Figure 2: Qualitative prefix and degeneration evidence assembled from four standalone subfigure files. In (a–b), each half shows the same examples across the prefix path; CoFiTok produces more organized intermediate denoising than endpoint-only factorization. In (c–d), shuffling breaks sample alignment, while deep synthesis can carry structure that the restricted, zero-preserving operator cannot. Quantitative path and zero-token results are reported in Tables 1 and 4.

Endpoint quality and prefixes. At the same 20k training budget, K8 light CoFiTok nearly matches the endpoint-only factorized control in endpoint x_0 MSE at $t = 500$ while greatly improving path alignment.

Across the four paired dataset-seed comparisons, CoFiTok lowers path AUC in 4/4 cases, with a mean relative reduction of 96.18%. This structural gain has a measurable endpoint tradeoff: endpoint x_0 MSE increases by 4.20% on average and is not lower in any of the four pairs. Figure 2 shows the corresponding qualitative prefix grids. In the full-validation deterministic sweep, K8 light also improves MSE AUC and low-res Frechet proxy AUC on both Tiny ImageNet-200 and ImageNet-64 HF, while the endpoint-only factorized control keeps the lower endpoint MSE.

Table 2: Matched 20k two-seed comparison (mean $\pm$ sample standard deviation over seeds 103 and 139). Endpoint x_0 MSE and path AUC use 1,024 validation images at $t = 500$; lower is better. Higher is better for effective token count.

Dataset	Method	Endpoint x_0 MSE $\downarrow$	Path AUC $\downarrow$	Eff. K $\uparrow$	Zero ratio $\downarrow$
Tiny ImageNet-200	Endpoint-only factorized	0.1253 ± 0.0001	1.0098 ± 0.1430	5.312 ± 0.049	0.0000 ± 0.0000
Tiny ImageNet-200	CoFiTok	0.1296 ± 0.0015	0.0378 ± 0.0007	6.766 ± 0.015	0.0000 ± 0.0000
ImageNet-64 HF	Endpoint-only factorized	0.1096 ± 0.0006	0.9984 ± 0.1210	5.169 ± 0.108	0.0000 ± 0.0000
ImageNet-64 HF	CoFiTok	0.1151 ± 0.0016	0.0379 ± 0.0037	6.752 ± 0.012	0.0000 ± 0.0000

Table 3: ImageNet-256 K4 20k confirmatory order and endpoint results over 1,024 validation images at $t = 500$.

Seed	E path	C ordered	non-ID mean	reverse	rank/24	Δ CI low	D endpoint	C endpoint
103	1.7879	0.0664	0.3157	0.7192	1	0.2489	0.1234	0.1133
139	1.4401	0.0678	0.3082	0.6821	1	0.2400	0.1244	0.1179

Learned order and ImageNet-256 scaling. Order ablations keep the same component sum at the endpoint but change the prefix trajectory. This makes order evaluation a direct test of whether prefixes are organized rather than an endpoint-quality test.

Table 3 gives the preregistered ImageNet-256 K4 20k two-seed test. Besides a fixed random permutation and reverse order, it exhaustively evaluates all 24 component permutations over 1,024 images and reports a paired-bootstrap confidence interval for the mean of the 23 non-identity orders. Because every order sums the same components, the endpoint is preserved; the test isolates the organization of the prefix path.

The preregistered confirmatory test passes all seven gates. For both seeds, the learned order ranks 1st of 24, every non-identity permutation has worse path AUC, and the paired-bootstrap lower bounds for non-identity minus ordered AUC are positive (0.2489 and 0.2400). The endpoint component sum is invariant up to 3×10^{-15} MSE. Relative to the parameter-matched direct dense control, CoFiTok also lowers endpoint x_0 MSE by 6.73% on average in this ImageNet-256 confirmatory setting.

Generation and related-method context. Cross-batch validation confirms that endpoint-only factorization remains a strong endpoint baseline. Generated-sample metrics do not establish stable unconditional superiority: the matched controls favor endpoint-only on Tiny ImageNet-200 and ImageNet-64 HF under the internal streamed protocol, and EDM or direct dense wins the broad Frechet-style rows. Official D-AR/MAR/ReTok 50K and FlexTok/TiTok reconstruction results remain secondary protocol tiers in the appendix; none is used as a matched CoFiTok superiority claim.

4.3 ABLATIONS

Matched component protocol. We isolate the K8 method components on ImageNet-64 HF with identical 5k-step training, seeds 103/139, and 1,024-image evaluation at $t = 500$. The full model uses $(\lambda_p, \lambda_c, p) = (0.15, 0.30, 1.5)$. Endpoint-only removes both path terms; the next two rows remove one term at a time; simultaneous prediction removes token feedback; deep synthesis replaces restricted S_k with a biased nonlinear decoder. Table 4 reports mean $\pm$ sample standard deviation.

Objective, prediction, and synthesis components. Relative to endpoint-only factorization, the full objective lowers mean path AUC by 93.0% with a 2.0% endpoint-MSE cost. Relative to removing only the prefix or component path term, it lowers path AUC by 45.7% and 37.9%, respectively, while endpoint MSE changes by less than 0.2%. Both path terms therefore improve the claimed interface rather than endpoint fit. Sequential feedback is a modest implementation gain, not a core requirement: full improves the two-seed mean by 2.8% in path AUC and 1.0% in endpoint MSE over simultaneous heads, but wins path AUC on only one seed. Deep synthesis attains lower path AUC yet has zero ratio 0.0485 ± 0.0050 versus exactly zero for restricted S_k ; its apparent gain violates the identifiability contract. Figure 2 visualizes zero/shuffle/deep- S_k diagnostics.

Table 4: Matched two-seed component ablations on ImageNet-64 HF. Lower is better for endpoint MSE, path AUC, and zero ratio; higher effective K indicates broader component use. All path values use the same progress-1.5 target.

Variant	Endpoint x_0 MSE $\downarrow$	Path AUC $\downarrow$	Eff. K $\uparrow$	Zero ratio $\downarrow$
CoFiTok (full)	0.1300 ± 0.0027	0.0686 ± 0.0017	6.780 ± 0.015	0.0000 ± 0.0000
Endpoint-only factorized	0.1275 ± 0.0019	0.9991 ± 0.1607	5.366 ± 0.054	0.0000 ± 0.0000
No path-prefix term	0.1300 ± 0.0003	0.1269 ± 0.0096	6.547 ± 0.027	0.0000 ± 0.0000
No path-component term	0.1302 ± 0.0012	0.1105 ± 0.0013	6.752 ± 0.069	0.0000 ± 0.0000
Simultaneous tokens	0.1314 ± 0.0016	0.0709 ± 0.0081	6.829 ± 0.025	0.0000 ± 0.0000
Deep synthesis	0.1496 ± 0.0305	0.0463 ± 0.0051	6.774 ± 0.032	0.0485 ± 0.0050

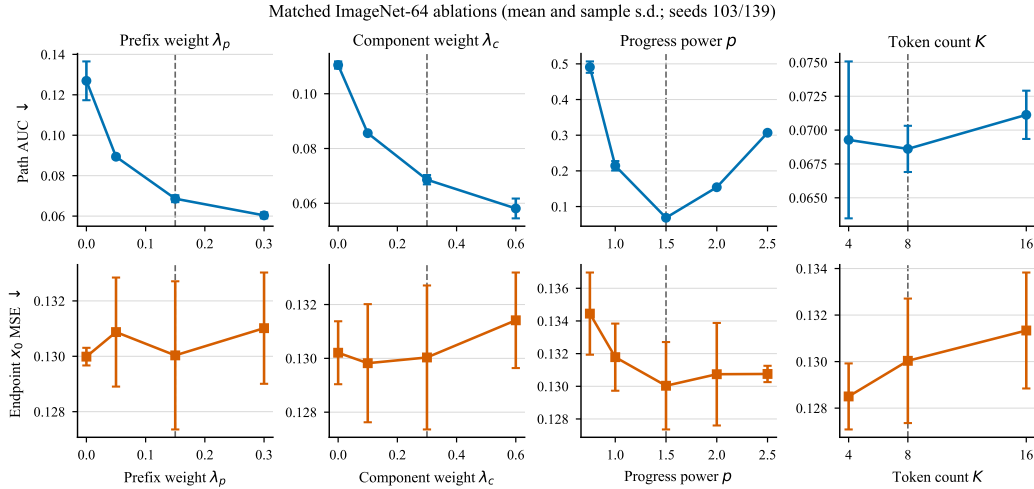


Figure 3: Matched ImageNet-64 HF hyperparameter ablations. Top: denoise-path AUC; bottom: endpoint x_0 MSE. Curves vary prefix weight λ_p , component weight λ_c , progress power p , and token count K . Lower is better; bars show sample standard deviation over seeds 103/139.

Hyperparameter sensitivity. Figure 3 varies the two path-loss weights, progress power, and token count. Every point is a two-seed mean with sample standard-deviation bars; the dashed line marks the default. Crucially, the power sweep is evaluated against one common progress-1.5 path rather than allowing each training target to define its own metric. Raising either path weight steadily improves path AUC with little endpoint movement; (0.15, 0.30) is a moderate tradeoff rather than the path-only optimum. Power 1.5 is the clear minimum under the common target. K8 gives the lowest mean path AUC, while K4 has slightly lower endpoint MSE and K16 adds cost without a 5k-step path gain.

Component-separation stress test. We treat component decorrelation as a pressure-term ablation rather than a default objective. The headline CoFiTok rows use the no-decorrelation light denoise-path objective because it most directly measures ordered prefix denoising. On Tiny ImageNet-200, a conservative schedule (weight=0.003, start step 3000, warmup 1500, full weight at step 4500 of 5000) reduces component correlation on two seeds while keeping the ordered-prefix penalty small.

Across the two seeds, scheduled decor 0.003 changes ordered path AUC by +0.0036, mean absolute component cosine by -0.0328, final MSE by -0.0002, and effective token count by +0.0336. Thus mild separation pressure can make components less correlated without materially breaking ordered prefix behavior. Stronger decor weights reduce cosine further but impose larger path-AUC penalties, so we keep them as stress-test ablations rather than the default CoFiTok objective.

5 DISCUSSION

CoFiTok suggests that dense pixel-space diffusion noise prediction can be treated as a sequence of ordered denoising-token components. The restricted synthesis operator is central: it gives the prefix interface meaning and prevents the method from becoming a generic latent decoder. The denoise-path objective is also central because early prefixes should represent partial denoising rather than clean reconstructions.

The strongest empirical result is the all-dataset prefix-control diagnostic: CoFiTok has lower cross-batch path AUC than the endpoint-only factorized control on all eight datasets, while matched objective, prediction, order, and deep- S_k ablations identify the roles of path supervision and restricted synthesis. The most important negative result is that generated-sample Frechet-style metrics do not show a stable unconditional generation-quality win. Together, these results define the current scope of the contribution.

6 LIMITATIONS

The current models are small and trained with short MVP budgets. Frechet-style sample-quality results are mixed and do not show a robust CoFiTok generation win; EDM is stronger on most generation rows, while direct dense prediction provides the strict factorization control. P1 FlexTok and TiTok rows use official pretrained reconstruction at 256x256 and are not retrained generation baselines. D-AR, MAR, and ReTok use separate official ImageNet-256 eval-only protocols and remain protocol-blocked under the matched-dataset/step training definition. Seed repeats cover the main endpoint-only/K8 comparison but not every ablation. The current manuscript is an ICLR-27 double-blind submission build. Sequential token prediction can also cost more latency and memory than a direct dense head; the current audit reports parameters, batch, nominal training images, and NFE, but not a standardized cross-implementation FLOP or wall-clock benchmark.

7 CONCLUSION

CoFiTok factorizes pixel-space diffusion noise prediction into an ordered sequence of denoising tokens. Each token is expanded by a restricted, condition-free synthesis operator into a dense negative-noise component, and prefix sums yield valid partial denoising results. Across the current eight-dataset evidence matrix, CoFiTok consistently improves prefix-control diagnostics over an endpoint-only factorized control and passes the restricted zero-token diagnostic where deep synthesis leaks token-independent structure. The current evidence therefore supports a scoped contribution about ordered restricted dense-noise factorization, while leaving raw unconditional generation quality to stronger monolithic diffusion baselines. Scaling to stronger backbones, longer training, and broader seed coverage remains future work.

REPRODUCIBILITY STATEMENT

The main paper specifies the token interface, restricted synthesis contract, training objective, datasets, evaluation protocols, and central metrics. The appendix provides the complete evidence matrix, source conditions, matched settings, ablations, and metric definitions. All reported tables are generated from the locked JSON evidence package described in the appendix, and the implementation and run records are maintained in the anonymous code package.

AI USE STATEMENT

Generative AI tools were used for code assistance, experiment-record organization, and language editing during project development. The authors reviewed and verified the implementation, mathematical formulation, data and evaluation protocols, numerical results, claims, and final manuscript. The authors take responsibility for the final content of this work and for all artifacts produced with this assistance.

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A SCOPE AND COVERAGE

This appendix separates comparison evidence from the main paper’s mechanism-centered tables by protocol, not by outcome. P0 contains matched-dataset and matched-step pixel-space training. P1 contains official-pretrained tokenizer reconstruction. A third tier contains official ImageNet-256 50K evaluations for architecture-incompatible methods. Values from different tiers are not used to claim direct superiority.

Table A1: Comparison protocols used throughout the paper and appendix.

Tier	Methods	Coverage	Evidence	Valid interpretation
P0 matched training	CoFiTok, direct dense, endpoint-only, ablations, Improved DDPM, EDM	Eight datasets	Train/eval under locked per-dataset steps	Within-row short-budget context; external methods are not compute matched
P1 reconstruction	FlexTok, TiTok	Eight datasets	Official-pretrained reconstruction on 256 images	Tokenizer reconstruction context only; not a generation comparison
Official related	D-AR, MAR, ReTok	ImageNet-256	Official-pretrained 50K evaluation	Secondary large-model context; not matched re-training against CoFiTok

B COMPLETE EVIDENCE COVERAGE

Table A2 exposes the full eight-dataset by 15-method Cartesian product. A blocked cell marks a cross-task comparison that is not interchangeable with the available official evaluation; official results are never counted as matched training.

Table A2: Evidence coverage matrix (8 datasets $\times$ 15 methods). T = completed matched train/eval; E = completed official-pretrained reconstruction eval-only; B = matched cross-task protocol blocked; O = separate official ImageNet-256 50K eval-only result. B+O retains the blocked matched cell while exposing the secondary official result.

Method	CIFAR	Tiny-IN	IN64-HF	IN64-strict	FFHQ	AFHQv2	IN256-10%	IN256
CoFiTok	T	T	T	T	T	T	T	T
Direct dense epsilon	T	T	T	T	T	T	T	T
Endpoint-only factorized	T	T	T	T	T	T	T	T
Legacy clean-prefix objective	T	T	T	T	T	T	T	T
No denoise-path prefix term	T	T	T	T	T	T	T	T
Add clean-space monotonic term	T	T	T	T	T	T	T	T
Simultaneous prediction	T	T	T	T	T	T	T	T
Deep- S_k synthesis	T	T	T	T	T	T	T	T
Improved DDPM	T	T	T	T	T	T	T	T
EDM	T	T	T	T	T	T	T	T
D-AR	B	B	B	B	B	B	B	B+O
FlexTok	E	E	E	E	E	E	E	E
TiTok	E	E	E	E	E	E	E	E
MAR	B	B	B	B	B	B	B	B+O
ReTok	B	B	B	B	B	B	B	B+O

The locked matrix contains 80 completed P0 train/eval cells, 16 P1 eval-only cells, and 24 protocol-blocked cells. Three official 50K rows remain separate.

C EXTERNAL-METHOD RESULTS

The next four tables are generated from the locked evidence JSON. They report P0 generation, FlexTok/TiTok reconstruction, official D-AR/MAR/ReTok ImageNet-256 results, and the paper’s claim boundary.

D HOW TO READ THE EXTERNAL COMPARISONS

Improved DDPM and EDM. Table A3 is the direct external pixel-diffusion context. Dataset, resolution, split, and nominal optimizer steps are aligned, but parameter count, batch size, nominal images seen, and NFE remain explicit. EDM is the low-resolution Frechet-style winner on seven of eight rows, and the direct dense predictor wins the remaining row. These results rule out a broad unconditional-generation superiority claim for CoFiTok. They do not test the paper’s distinctive interface claim because monolithic predictors do not expose ordered component prefixes.

Table A3: Eight-dataset P0 generated-sample comparison under the locked short-budget protocols. Lower Frechet-style metrics are better.

Dataset	C low	D low	DDPM low	EDM low	low winner	C Inc	D Inc	DDPM Inc	EDM Inc	Inc winner	C rank
CIFAR-10	6.47	5.80	7.86	2.87	EDM	312.00	359.50	498.65	351.17	CoFiTok	1
Tiny-IN	8.49	6.99	9.07	3.10	EDM	373.63	392.40	381.24	268.72	EDM	2
ImageNet-64 HF	8.09	6.89	8.88	3.23	EDM	390.99	417.79	378.33	258.40	EDM	3
ImageNet-64 strict	8.36	6.87	8.80	2.75	EDM	374.68	383.72	355.11	248.31	EDM	3
FFHQ-64	8.74	6.84	9.44	2.51	EDM	425.71	427.05	369.01	325.49	EDM	3
AFHQv2-64	7.08	5.68	8.29	1.96	EDM	331.06	333.42	335.18	230.40	EDM	2
ImageNet-256 10%	10.44	12.59	10.40	10.39	EDM	414.19	366.42	419.60	368.68	Direct dense epsilon	3
ImageNet-256	10.54	10.24	10.40	10.40	Direct dense epsilon	417.26	354.13	414.30	368.43	Direct dense epsilon	4

Table A4: *

C = CoFiTok; D = parameter-matched direct dense epsilon. Rows use the locked per-dataset short-budget protocol; they are not formal SOTA FID claims.

Table A5: Official-pretrained tokenizer reconstruction context (eval-only).

Dataset	src	eval	FlexTok PSNR	FlexTok low	TiTok PSNR	TiTok low
AFHQv2-64	64	256	24.04	0.036	17.29	0.217
CIFAR-10	32	256	26.77	0.037	19.35	0.168
ImageNet-64 strict	64	256	22.57	0.042	16.73	0.196
FFHQ-64	64	256	24.06	0.042	17.39	0.231
ImageNet-64 HF	64	256	23.60	0.039	18.04	0.182
ImageNet-256	256	256	19.12	0.044	15.54	0.196
ImageNet-256 10%	256	256	19.12	0.044	15.54	0.196
Tiny-IN	64	256	21.81	0.044	17.30	0.160

Table A6: *

Official pretrained reconstruction at 256x256 on 256 images per dataset. These rows are not retrained generation baselines.

FlexTok and TiTok. Table A5 uses official-pretrained models and evaluates reconstruction at 256×256 on 256 images per dataset. These methods tokenize image content; CoFiTok factorizes a dense diffusion noise prediction. Their reconstruction PSNR and Frechet-style values therefore document adjacent capability rather than a shared generation objective.

D-AR, MAR, and ReTok. Table A7 reports 50K-sample official evaluations on ImageNet-256. The rows use available pretrained checkpoints and their corresponding generation protocols. They are useful large-model context, but they cannot be relabeled as eight-dataset matched-step retraining. The Inception Score for MAR follows the ADM-style evaluator used in the recorded run and is not mixed with the differently scaled value reported by another evaluation implementation.

E MATCHED P0 PROTOCOL DETAILS

The P0 protocol fixes dataset identity, spatial resolution, split, and nominal optimizer steps. It does not pretend that different published architectures have equal FLOPs. Table A10 exposes the remaining major differences instead of hiding them in a single rank.

Only the direct dense control isolates factorization at nearly equal parameter count. Improved DDPM and EDM test whether the short-budget generator is competitive with established monolithic objectives, while endpoint-only factorized prediction tests whether ordered prefix supervision contributes beyond retaining K token components.

F DATASET AND SOURCE CONDITIONS

The two ImageNet-64 aliases are intentionally distinct. Results labeled ImageNet-64 HF use the HF-derived source; results labeled ImageNet-64 strict use the exact downsampled ImageNet source. No result is silently renamed between them.

Table A7: Official ImageNet-256 related-method evaluations (50K samples).

Method	Dataset	Protocol	Samples	FID	sFID	IS	Prec.	Rec.
D-AR	ImageNet-256	official 50K eval-only	50000	2.628	6.720	285.89	0.803	0.588
MAR	ImageNet-256	official 50K eval-only	50000	2.338	4.700	62.90	0.822	0.572
ReTok	ImageNet-256	official 50K eval-only	50000	2.219	5.896	245.94	0.816	0.599

Table A8: *

These rows use official pretrained protocols where available and are secondary comparisons; they are not part of the matched-dataset/optimizer-step P0 matrix.

Table A9: Claim-evidence boundary.

Claim axis	Evidence summary
Generation quality	CoFiTok is best on 0/8 lowres rows and 1/8 Inception rows; average ranks are 3.12 and 2.62. Do not claim generation SOTA.
Prefix control	CoFiTok path AUC is lower than the endpoint-only factorized control on 8/8 datasets.
Repeated 20k evidence	CoFiTok lowers path AUC in 4/4 paired dataset-seed runs (mean relative reduction 96.18%), while endpoint x_0 MSE at $t = 500$ changes by +4.20% on average.
ImageNet-256 scaling	ImageNet-256 K4 20k confirmatory gates: 7/7 passed; overall PASS; mean endpoint-MSE change -6.73%; ordered ranks among all 24 permutations 1/1.
Restricted synthesis	CoFiTok zero-token ratio is zero in current rows; deep- S_k has nonzero zero-token ratio on 8/8 datasets.
Related methods	completed official 50K: D-AR, MAR, ReTok. All rows remain secondary and outside P0 matched-dataset/step training.
Main paper stance	The core empirical gates support the scoped claim of ordered restricted dense-noise factorization with prefix-controllable denoising, not broad unconditional generation superiority.

G METRIC DEFINITIONS

For prefix budget m , the cumulative prediction and pixel-space estimate are

$$\hat{\epsilon}^{(1:m)} = \sum_{k=1}^m S_k(z^{(k)}), \quad \hat{x}_0^{(m)} = \frac{x_t - \sigma_t \hat{\epsilon}^{(1:m)}}{\alpha_t}. \quad (10)$$

Let e_m be the mean-squared error between $\hat{x}_0^{(m)}$ and the preregistered denoise-path target at prefix m . The reported path AUC is the equal-spaced trapezoid average

$$\text{AUC}_{\text{path}} = \frac{1}{K-1} \sum_{m=1}^{K-1} \frac{e_m + e_{m+1}}{2}. \quad (11)$$

Lower values indicate that intermediate prefixes follow the intended partial denoising path. Endpoint MSE instead uses only $m = K$ and does not measure the quality of intermediate prefixes.

For component energy proportions p_k , effective token count is

$$K_{\text{eff}} = \exp\left(-\sum_k p_k \log p_k\right). \quad (12)$$

The zero-token ratio divides the energy produced by $S_k(0)$ by the energy of the corresponding observed components. A bias-free restricted synthesis operator satisfies this diagnostic exactly up to numerical precision.

H REPEATED AND CONFIRMATORY EVIDENCE

The matched 20K repeat uses K8 models, seeds 103 and 139, and 1,024 validation images at $t = 500$ on Tiny ImageNet-200 and ImageNet-64 HF. CoFiTok lowers path AUC in all four paired dataset-seed comparisons, with a mean relative reduction of 96.18%. The mean endpoint x_0 -MSE change is +4.20%, so this repeat supports prefix organization rather than an endpoint-quality win.

The ImageNet-256 K4 20K confirmatory run uses the same two seeds and exhaustively evaluates all 24 component permutations. The learned order ranks first for both seeds, the paired-bootstrap

Table A10: Locked P0 short-budget settings. C = CoFiTok; D = direct dense; E = endpoint-only factorized. NFE is listed in the order C/D/Improved-DDPM/EDM.

Protocol family	Datasets	Resolution	Steps	C/D/E batch	DDPM/EDM batch	Quality images	Sample/real images	NFE
CIFAR-10	CIFAR-10	32	3000	64	32	512	1024/4096	50/50/50/79
64px	Tiny-IN, IN64-HF, IN64-strict, FFHQ, AFHQv2	64	5000	32	32	512	1024/4096	50/50/50/79
ImageNet-256	10% and full train	256	5000	4	4	256	512/2048	50/50/50/79

Table A11: Trainable parameter counts in the P0 protocols. The direct dense predictor is the parameter-matched factorization control; Improved DDPM and EDM are external architecture base-lines.

Resolution	CoFiTok	Direct dense	Dense delta vs. CoFiTok	Improved DDPM	EDM
32 × 32	81,808	80,707	−1.35%	9,593,539	3,870,531
64 × 64	81,808	80,707	−1.35%	21,715,907	3,870,531
256 × 256	15,776	15,690	−0.55%	7,200,579	952,291

lower bounds for non-identity minus ordered AUC are positive, and every permutation preserves the endpoint component sum up to 3×10^{-15} MSE. Relative to the parameter-matched direct dense control, CoFiTok lowers endpoint MSE by 6.73% on average in this confirmatory setting.

I MATCHED COMPONENT ABLATIONS

The component ablation suite uses ImageNet-64 HF, K8, 5,000 optimizer steps, seeds 103/139, and 1,024 validation images at fixed $t = 500$. All variants share the data, model width, optimizer, and evaluation noise for each seed. The progress-power sweep is scored against one common $p = 1.5$ path; it is not scored against a different target for each point.

The direct removals distinguish the path-prefix and path-component terms from the endpoint objective. The simultaneous variant removes token-to-token feedback while retaining the same heads and losses. The deep-synthesis variant retains token-only inputs but introduces biased nonlinear layers, so its zero-token ratio directly tests whether decoder capacity can carry a token-independent prior. Full CoFiTok lowers path AUC by 93.0% relative to endpoint-only and by 45.7%/37.9% relative to removing the prefix/component path term, respectively. Endpoint MSE changes by less than 0.2% in the two single-term comparisons. Sequential feedback changes the mean more modestly (−2.8% path AUC and −1.0% endpoint MSE) and wins path AUC on only one of two seeds, so it is treated as an optional predictor implementation rather than part of the core claim. Deep synthesis reaches lower path AUC but produces a nonzero zero-token ratio of 0.0485 ± 0.0050 . The parameter-matched direct dense control is only 0.6% lower in endpoint MSE at 64px but exposes no multi-prefix interface; conversely, the ImageNet-256 confirmatory setting gives CoFiTok a 6.73% endpoint advantage over direct dense. Factorization is therefore supported as an interface with a small or favorable endpoint tradeoff, not as a universal endpoint-quality guarantee.

J HYPERPARAMETER SENSITIVITY

Figure A1 and Table A15 report the full two-seed sweeps. The dashed lines mark the main values: $\lambda_p = 0.15$, $\lambda_c = 0.30$, $p = 1.5$, and $K = 8$. Increasing either loss weight improves path AUC with little endpoint movement; the chosen weights are a moderate endpoint/path tradeoff. Power 1.5 is the minimum under the common evaluation target. K8 has the lowest mean path AUC, K4 the lowest endpoint MSE, and K16 no short-budget path advantage.

Table A16: Exploratory ImageNet-256 confirmatory generation controls.

These rows do not replace the path and endpoint preregistered tests. Lower is better.

Table A12: Dataset roles in the locked evidence matrix.

Dataset label	Resolution	Split role	Source condition
CIFAR-10	32	Train/test	Canonical CIFAR-10 staging
Tiny-IN	64	Train/validation	Tiny ImageNet-200 staging
ImageNet-64 HF	64	Train/validation	HF-derived ImageNet-1K 64x64 source
ImageNet-64 strict	64	Train/validation	Exact downsampled ImageNet source from Academic Torrents
FFHQ-64	64	Train/evaluation	Non-classification face-domain extension
AFHQv2-64	64	Train/evaluation	Non-classification animal-domain extension
ImageNet-256 10%	256	10% train/full validation	Deterministic subset derived from the staged 256x256 source
ImageNet-256	256	Full train/validation	Full staged 256x256 ImageNet source

Table A13: Matched component ablations (mean $\pm$ sample standard deviation). Lower is better for endpoint MSE, path AUC, and zero ratio; higher effective K indicates broader component use.

Variant	Endpoint x_0 MSE $\downarrow$	Path AUC $\downarrow$	Eff. K $\uparrow$	Zero ratio $\downarrow$
CoFiTok (full)	0.1300 $\pm$ 0.0027	0.0686 $\pm$ 0.0017	6.780 $\pm$ 0.015	0.0000 $\pm$ 0.0000
Endpoint-only factorized	0.1275 $\pm$ 0.0019	0.9991 $\pm$ 0.1607	5.366 $\pm$ 0.054	0.0000 $\pm$ 0.0000
No path-prefix term	0.1300 $\pm$ 0.0003	0.1269 $\pm$ 0.0096	6.547 $\pm$ 0.027	0.0000 $\pm$ 0.0000
No path-component term	0.1302 $\pm$ 0.0012	0.1105 $\pm$ 0.0013	6.752 $\pm$ 0.069	0.0000 $\pm$ 0.0000
Simultaneous tokens	0.1314 $\pm$ 0.0016	0.0709 $\pm$ 0.0081	6.829 $\pm$ 0.025	0.0000 $\pm$ 0.0000
Deep synthesis	0.1496 $\pm$ 0.0305	0.0463 $\pm$ 0.0051	6.774 $\pm$ 0.032	0.0485 $\pm$ 0.0050

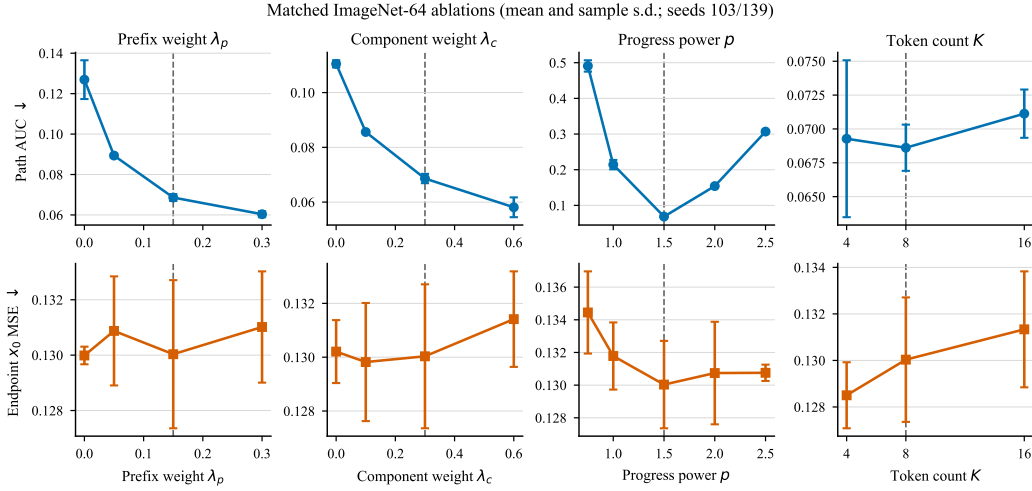
Seed	Method	Samples	Lowres Frechet	Inception Frechet
103	Endpoint-only factorized	4096	11.1619	370.3672
103	Direct dense	4096	9.7396	449.3913
103	CoFiTok	4096	10.8232	349.4076
139	Endpoint-only factorized	4096	8.9753	404.6116
139	Direct dense	4096	10.5597	384.8126
139	CoFiTok	4096	11.0839	356.9311

K EVIDENCE LOCK AND REPRODUCIBILITY

All numerical tables imported above are generated from the final locked JSON evidence package. The paper source does not hand-edit those metrics. The locked state contains the per-run configuration, dataset alias, split, optimizer steps, batch size, seed, parameter count, NFE, evaluation image count, and sample count needed to distinguish the three protocol tiers. New experiments must produce a new evidence package rather than silently overwrite these rows.

Table A14: Parameter-matched architecture control under the same two-seed 5k ImageNet-64 HF protocol. Direct dense has no native multi-prefix path.

Variant	Endpoint x_0 MSE $\downarrow$	Path AUC $\downarrow$	Eff. K $\uparrow$	Zero ratio $\downarrow$
CoFiTok (full)	0.1300 ± 0.0027	0.0686 ± 0.0017	6.780 ± 0.015	0.0000 ± 0.0000
Parameter-matched direct dense	0.1292 ± 0.0006	—	1.000 ± 0.000	0.0000 ± 0.0000

Figure A1: Matched ImageNet-64 HF hyperparameter sensitivity. Top row: denoise-path AUC; bottom row: endpoint x_0 MSE. Error bars are sample standard deviations over seeds 103/139. Lower is better.Table A15: Numerical values underlying Figure A1. Each entry is mean $\pm$ sample standard deviation over seeds 103/139.

Axis	Value	Endpoint x_0 MSE $\downarrow$	Path AUC $\downarrow$	Eff. K $\uparrow$
λ_p	0	0.1300 ± 0.0003	0.1269 ± 0.0096	6.547 ± 0.027
	0.05	0.1309 ± 0.0020	0.0894 ± 0.0009	6.694 ± 0.033
	0.15	0.1300 ± 0.0027	0.0686 ± 0.0017	6.780 ± 0.015
	0.3	0.1310 ± 0.0020	0.0604 ± 0.0015	6.819 ± 0.002
λ_c	0	0.1302 ± 0.0012	0.1105 ± 0.0013	6.752 ± 0.069
	0.1	0.1298 ± 0.0022	0.0856 ± 0.0004	6.860 ± 0.001
	0.3	0.1300 ± 0.0027	0.0686 ± 0.0017	6.780 ± 0.015
	0.6	0.1314 ± 0.0018	0.0581 ± 0.0036	6.717 ± 0.010
p	0.75	0.1344 ± 0.0025	0.4909 ± 0.0159	7.035 ± 0.102
	1	0.1318 ± 0.0021	0.2143 ± 0.0133	7.912 ± 0.015
	1.5	0.1300 ± 0.0027	0.0686 ± 0.0017	6.780 ± 0.015
	2	0.1307 ± 0.0031	0.1541 ± 0.0010	5.449 ± 0.020
	2.5	0.1308 ± 0.0005	0.3069 ± 0.0031	4.459 ± 0.007
K	4	0.1285 ± 0.0014	0.0693 ± 0.0058	3.321 ± 0.006
	8	0.1300 ± 0.0027	0.0686 ± 0.0017	6.780 ± 0.015
	16	0.1313 ± 0.0025	0.0711 ± 0.0018	13.676 ± 0.154